

Learning and Estimation of Latent Structural Models Based on between-Data Metrics

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Abstract—With the development of information technology, a wide variety of data have been accumulated, and there are many methods for analyzing such data. In this study, we model the input data and the metrics between the data based on the assumption that each metric is generated from a continuous latent variable. Specifically, we assume that the input data are generated using low-dimensional latent variables and their projection matrices. We describe a method for estimating the latent variables. Because the generative model defined in this study cannot obtain the Q function analytically, we use the Monte Carlo EM algorithm to approximate the Q function and investigate an efficient parameter estimation method. Experiments using artificial data and the 20 newsgroups dataset demonstrate the effectiveness of the proposed method.

Index Terms—continuous latent variable, Monte Carlo EM algorithm, dimensionality reduction

I. INTRODUCTION

With the development of information technology, various analysis methods for diverse data have been proposed. Examples include the k -NN and k -means methods, which assume a metric between the data [1].

Several fields have focused on such a metric, of which there are two main categories:

- (1) metrics used to obtain a better representation of the relationship between the input data and
- (2) metrics to extract the relationship between the input data based on a pre-existing metric.

Research related to the former is called distance metric learning [2] – [4]. Distance metric learning refers to a method for learning a distance structure from the training data, taking its statistical properties into account under arbitrary constraints. For the latter, there is a method called statistical relational learning [5], [6], which expresses the relationships between objects such as users and home pages as metrics and extracts the relationships between the data.

Because distance metric learning estimates an appropriate distance measure from the training data, it can be interpreted that statistical relational learning is a method for extracting objects with strong relationships.

By contrast, methods such as a factor analysis [7] and a probabilistic principal component analysis [8], which explicitly assume continuous latent variables in the input data, can model the data generation process. Using such continuous latent variables, samples of the input data can be generated. However, few studies have modeled the process for generating the metrics. With the ability to model this

process, various applications such as knowledge discovery from data and clarifying the latent structure behind the data are possible.

Based on the above discussion, this study focuses on such a latent variable and assumes that the metrics between the data are obtained from the inner product. In addition, assuming that each latent variable follows a certain probability distribution, we model its generative process and propose a parameter estimation method for the defined model. Moreover, we also propose a method for achieving an efficient parameter estimation. The effectiveness of the proposed method is demonstrated through experiments using both artificial data and benchmark datasets.

II. PRELIMINARIES

A. Proposed Generative Model

Let the input data be $\mathbf{x}_i \in \mathcal{R}^d, i = 1, 2, \dots, n$, and the set of $\mathbf{x}_i$ be $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$. Suppose that each $\mathbf{x}_i$ is generated from the continuous latent variable $\mathbf{z}_i \in \mathcal{R}^k, k \ll d$, and the set of latent variables $\mathbf{z}_i$ is $\mathcal{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n\}$. Here, we assume that $\mathbf{z}_i$ follows $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \sigma_z^2 \mathbf{I})$.

The matrix used to map the latent variable $\mathbf{z}_i$ to the space of input data is defined as $\mathbf{F} \in \mathcal{R}^{d \times k}$, with $\boldsymbol{\mu}$ for the mean term and $\boldsymbol{\varepsilon}_i$ for the noise term. Herein, we assume that $\mathbf{x}_i$ is generated by

$$\mathbf{x}_i = \mathbf{F} \mathbf{z}_i + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_i, \quad (1)$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ and $\boldsymbol{\Sigma}$ is a diagonal matrix whose elements are $\sigma_m^2 (m = 1, 2, \dots, d)$.

Next, let the metric between input data $\mathbf{x}_i, \mathbf{x}_j$ be y_{ij} , and assume that the metric is obtained by taking the inner product of the latent variables $\mathbf{z}_i, \mathbf{z}_j$. Here, letting the set of y_{ij} be $\mathcal{Y} = \{y_{12}, y_{13}, \dots, y_{n-1, n}\}$, and the noise term of y_{ij} be e_{ij} , we assume that y_{ij} is generated from

$$y_{ij} = \mathbf{z}_i^T \mathbf{z}_j + e_{ij}, \quad (2)$$

where, e_{ij} follows $e_{ij} \sim \mathcal{N}(0, \sigma_y^2)$.

Fig. 1 illustrates our proposed graphical model.

B. Monte Carlo EM Algorithm

The EM algorithm generally implements a parameter estimation using latent class models. However, depending on the model assumed, the update equation in the EM algorithm, specifically the E-step, may not be analytically feasible. For

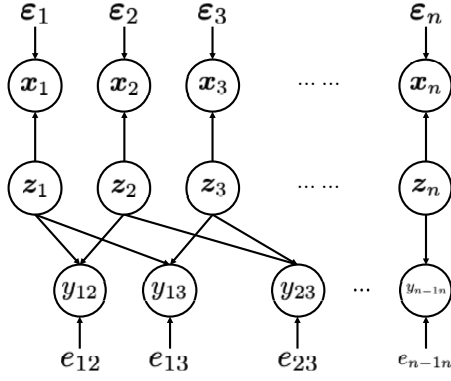


Fig. 1: Graphical model of the proposed method

such models, the Monte Carlo EM algorithm can be used to approximate the E-step based on sampling [9].

In the original EM algorithm, letting the parameter vector before the update be θ^{old} , the parameter vector be θ , the latent variable be $\mathbf{Z}$, and the observed variable be $\mathbf{X}$, Q function is then defined as follows:

$$Q(\theta, \theta^{old}) = \int p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{Z}, \mathbf{X}|\theta) d\mathbf{Z}. \quad (3)$$

By contrast, the Monte Carlo EM algorithm approximates the integral of the right-hand side of Eq. (3) by a finite sum of L sets of samples $\{\mathbf{Z}^{(l)}\}$ generate from the current estimate $p(\mathbf{Z}|\mathbf{X}, \theta^{old})$ as follows:

$$Q(\theta, \theta^{old}) \simeq \frac{1}{L} \sum_{l=1}^L \ln p(\mathbf{Z}^{(l)}, \mathbf{X}|\theta). \quad (4)$$

Here, the same method as in the original EM algorithm is used to maximize the Q function during M-step.

C. Related research

As mentioned before, several studies focus on the metric between the input data, such as distance metric learning and statistical relational learning.

Among these, distance metric learning assumes a Mahalanobis distance between the input data and learns the distance structure, *i.e.*, the Mahalanobis matrix, under arbitrary constraints [2]. As an example of such constraints, if the data belongs to the same category, the distance should be smaller than or equal to the threshold value while maintaining the distance between the data belonging to different categories.

By contrast, in statistical relational learning, a metric indicating the relationship between the data is given by a particular measure deterministically [5]. This is a method for extracting the relationships between the input data, given a metric indicating the relationship between the data. There are a variety of methods for statistical relational learning including spectral clustering [10] and infinite relation models [11]. Statistical relational learning methods such as a probabilistic relational PCA [12] assume a latent structure behind the input data. However, with such method, the metric between the data is given deterministically, and its generation is not modeled.

Among these, distance metric learning is a method used to provide a distance measure that considers the statistical

properties of the input data, and statistical relational learning is a method for simply extracting the relationships behind the data obtained. This indicates that this methodology is unable to handle the problem setting in which a metric is generated from latent variables, such as the problem applied in the present study.

III. PROPOSED METHOD

A. Definition of proposed model and applying Monte Carlo EM algorithm

From the assumptions above, the likelihood function L is defined as follows:

$$L = \prod_{i=1}^n p(\mathbf{x}_i|F, \mathbf{z}_i, \boldsymbol{\mu}, \boldsymbol{\Sigma}) \left\{ \prod_{i \neq j}^n p(y_{ij}|\mathbf{z}_i, \mathbf{z}_j, \sigma_y^2) \right\}^{\frac{1}{2}}. \quad (5)$$

In this study, we assume that the metric y_{ij} between the input data $\mathbf{x}_i, \mathbf{x}_j$ is obtained by taking the inner product of $\mathbf{z}_i, \mathbf{z}_j$ as described above. However, when applying the EM algorithm under these conditions, it is difficult to compute the integral in the Q function. For this reason, the Monte Carlo EM algorithm described above is introduced as a parameter estimation.

To implement the Monte Carlo EM algorithm, we need to obtain the posterior distribution $p(\mathbf{Z}|\mathbf{X}, \mathbf{Y}, \theta)$ for the latent variable $\mathbf{Z}$, and generate the data that follow it. Letting L groups of pairs of data generated by this posterior distribution be $\mathbf{Z}^{(l)} (l = 1, \dots, L)$, the Q function can be approximated as follows:

$$Q(\theta, \theta^{old}) \simeq \frac{1}{L} \sum_{l=1}^L \ln p(\mathbf{Z}^{(l)}, \mathbf{X}, \mathbf{Y}|\theta). \quad (6)$$

A parameter estimation is implemented by maximizing θ in Eq. (6).

B. Derivation of the posterior distribution for $\mathbf{z}_i$

To apply the Monte Carlo EM algorithm, it is necessary to compute the posterior distribution of $\mathbf{Z}$, and generates samples from it. Here, assuming that θ is the parameter vector, the posterior distribution $p(\mathbf{Z}|\mathbf{X}, \mathbf{Y}, \theta)$ for $\mathbf{Z}$ is defined as follows:

$$\begin{aligned} p(\mathbf{Z}|\mathbf{X}, \mathbf{Y}, \theta) &= \frac{p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}|\theta)}{\int p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}|\theta) d\mathbf{Z}} \\ &= \frac{p(\mathbf{Z})p(\mathbf{X}|\mathbf{Z}, \theta)p(\mathbf{Y}|\mathbf{Z})}{\int p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}|\theta) d\mathbf{Z}} \\ &= \frac{\prod_{i=1}^n \alpha_{ij}}{\int \prod_{i=1}^n \alpha_{ij} d\mathbf{z}_i}. \end{aligned} \quad (7)$$

To simplify the above equation, α_{ij} in Eq. (7) is defined as follows:

$$\alpha_{ij} = p(\mathbf{z}_i, \sigma_z^2 I) p(\mathbf{x}_i|\mathbf{z}_i, F, \boldsymbol{\mu}, \boldsymbol{\Sigma}) \prod_{j \neq i}^n p(y_{ij}|\mathbf{z}_i, \mathbf{z}_j, \sigma_y^2)^{\frac{1}{2}}. \quad (8)$$

However, it is difficult to calculate the posterior distribution of $\mathbf{Z}$ directly. For this reason, we apply Gibbs sampling to obtain the posterior distribution of the $\mathbf{z}_i$.

Summarizing Eq. (7) for $\mathbf{z}_i$, the posterior distribution for $\mathbf{z}_i$ can be obtained as $p(\mathbf{z}_i|\mathbf{x}_i, \mathbf{Y}, \mathbf{Z}_{\setminus i}, \theta) = \mathcal{N}(\mathbf{z}_i|\boldsymbol{\eta}_i, A_i)$.

Here, $\mathbf{Z}_{\setminus i}$ denotes the data set excluding $\mathbf{z}_i$ from the set of latent variables $\mathbf{Z}$, and $\boldsymbol{\eta}_i, A_i$ is defined as follows:

$$A_i = \left\{ \frac{1}{\sigma_z^2} I + \frac{1}{2\sigma_y^2} \sum_{j \neq i} \mathbf{z}_j \mathbf{z}_j^T + F^T \boldsymbol{\Sigma}^{-1} F \right\}, \quad (9)$$

$$\boldsymbol{\eta}_i = A_i^{-1} \left(F^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i + \boldsymbol{\mu}) + \frac{1}{\sigma_y^2} \sum_{j \neq i} y_{ij} \mathbf{z}_j \right) \quad (10)$$

For the derivation of $A_i, \boldsymbol{\eta}_i$, see the appendix.

C. Estimation of parameter $\boldsymbol{\theta}$

When using the Monte Carlo EM algorithm, the Q function of the proposed method is approximated through the following:

$$\begin{aligned} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{old}) &\simeq \frac{1}{L} \sum_{l=1}^L \left\{ \ln p(\mathbf{X} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) p(\mathbf{Y} | \mathbf{Z}^{(l)}) p(\mathbf{Z}^{(l)}) \right\} \\ &= \frac{1}{L} \sum_{l=1}^L \left\{ \ln p(\mathbf{X} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) + \ln p(\mathbf{Y} | \mathbf{Z}^{(l)}) \right. \\ &\quad \left. + \ln p(\mathbf{Z}^{(l)}) \right\}. \end{aligned} \quad (11)$$

Parameters that maximizes Eq. (11) can be derived as follows:

$$F = \left(\sum_{l=1}^L \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu}) \mathbf{z}_i^{(l)T} \right) \left(\sum_{l=1}^L \sum_{i=1}^n \mathbf{z}_i^{(l)} \mathbf{z}_i^{(l)T} \right)^{-1}, \quad (12)$$

$$\boldsymbol{\mu} = \frac{1}{Ln} \sum_{l=1}^L \sum_{i=1}^n (\mathbf{x}_i - F \mathbf{z}_i^{(l)}), \quad (13)$$

$$\begin{aligned} \boldsymbol{\Sigma} &= \text{diag} \left(\frac{1}{Ln} \sum_{l=1}^L \sum_{i=1}^n (\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}) \cdot \right. \\ &\quad \left. (\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu})^T \right), \end{aligned} \quad (14)$$

$$\sigma_y^2 = \frac{1}{Lnh} \sum_{l=1}^L \sum_{i=1}^n \sum_{j=1}^n \left(y_{ij} - \mathbf{z}_i^{(l)T} \mathbf{z}_j^{(l)} \right)^2, \quad (15)$$

$$\sigma_z^2 = \frac{1}{Lkn} \sum_{l=1}^L \sum_{i=1}^n \left\| \mathbf{z}_i^{(l)} \right\|^2. \quad (16)$$

Here, h is the number of elements in $\mathcal{Y}$, and diag is the operation that sets all non-diagonal elements of the matrix to zero.

From the above discussion, the computation of the posterior probabilities and the parameter estimation can be implemented through the following steps:

Step 1 To initialize parameters $\boldsymbol{\theta}$.

Step 2 Calculate $\boldsymbol{\eta}_i$ and A_i using Eqs. (9) and (10), sampling $\mathbf{z}_i$ by the posterior distribution $p(\mathbf{z}_i | \mathbf{x}_i, \mathbf{Y}, \mathbf{Z}_{\setminus i}, \boldsymbol{\theta})$.

Step 3 Repeat Step 2 for i , to obtain $\mathbf{Z}^{(l)}$.

Step 4 Repeat Step 3 for $l = 1, \dots, L$.

Step 5 Using the obtained $\mathbf{Z}^{(l)}$, calculate the parameters $\boldsymbol{\theta}$ using Eqs. (12) – (16).

Step 6 Repeat Step 2 – Step 5 until convergence.

D. Improving the efficiency of algorithms deriving posterior distribution

Although the algorithm described above can be used for a parameter estimation, the inverse of matrix A_i must be computed when calculating Eq. (9) and (10) in Step 2. In general, computing A_i^{-1} requires a computational cost of $O(k^3)$, and it is inefficient to compute it every time $\mathbf{z}_i^{(l)}$ is generated. In particular, when the number of dimensions of the input data significantly increases, the number of dimensions of the latent variable is also expected to be relatively large, so it is important to reduce the computational cost.

To reduce the computational cost, we focus on the second term on the right-hand side of Eq. (9). Here, letting A be the sum of A_i and $\frac{1}{2\sigma_y^2} \mathbf{z}_i \mathbf{z}_i^T$, then A can thus be defined as follows:

$$A = \frac{1}{\sigma_z^2} I + \frac{1}{2\sigma_y^2} \sum_{j=1}^n \mathbf{z}_j \mathbf{z}_j^T + F^T \boldsymbol{\Sigma}^{-1} F. \quad (17)$$

By applying Woodbury's formula [13] to Eq. (17), A_i^{-1} can be obtained by the following:

$$\begin{aligned} A_i^{-1} &= \left(A - \frac{1}{2\sigma_y^2} \mathbf{z}_i \mathbf{z}_i^T \right)^{-1} \\ &= A^{-1} - A^{-1} \mathbf{z}_i (-2\sigma_y^2 + \mathbf{z}_i^T A^{-1} \mathbf{z}_i) \mathbf{z}_i^T A^{-1} \end{aligned} \quad (18)$$

This makes it possible to calculate A_i^{-1} by simply calculating A^{-1} once, thus reducing the computational cost.

Furthermore, letting A^{new} be the updated value of A after implementing the algorithm, and $\mathbf{z}_i^{\text{new}}$ be the updated parameter of $\mathbf{z}_i$,

$$\begin{aligned} A^{\text{new}} &= A + \frac{1}{2\sigma_y^2} \left\{ \mathbf{z}_i^{\text{new}} (\mathbf{z}_i^{\text{new}})^T - \mathbf{z}_i \mathbf{z}_i^T \right\} \\ &= A_i + \frac{1}{2\sigma_y^2} \mathbf{z}_i \mathbf{z}_i^T + \frac{1}{2\sigma_y^2} \left\{ \mathbf{z}_i^{\text{new}} (\mathbf{z}_i^{\text{new}})^T - \mathbf{z}_i \mathbf{z}_i^T \right\} \\ &= A_i + \frac{1}{2\sigma_y^2} \mathbf{z}_i^{\text{new}} (\mathbf{z}_i^{\text{new}})^T, \end{aligned} \quad (19)$$

are respectively obtained.

By using Woodbury's formula for the obtained A_i^{-1} ,

$$\begin{aligned} (A^{\text{new}})^{-1} &= A_i^{-1} \\ &\quad - A_i^{-1} \mathbf{z}_i^{\text{new}} \left(2\sigma_y^2 + (\mathbf{z}_i^{\text{new}})^T A_i^{-1} \mathbf{z}_i^{\text{new}} \right) (\mathbf{z}_i^{\text{new}})^T A_i^{-1}, \end{aligned} \quad (20)$$

is derived. Equations (19) and (20) show that the cost of computing the posterior distribution parameters can be reduced by applying the updated versions of A_i and $\mathbf{z}_i$.

IV. EXPERIMENT WITH ARTIFICIAL DATA SETS

A. Experimental conditions

To demonstrate that the proposed method can correctly estimate the latent variables and parameters, we conducted an experiment using an artificial dataset. For this experiment, we set the true parameters in advance and examined the difference between the estimated parameters and their convergence.

The dimensionality of the data used is $d = 5$, the dimensionality of the latent variable $\mathbf{z}_i$ is $k = 3$, the number

of data is $n = 100$, and the numbers of data pairs L for approximating the Q function are set to 50, 100 and 300, respectively. In addition, the true parameters are set to $\sigma_z^{2*} = 0.1$, $\sigma_y^{2*} = 1$, and the components of F^* , μ^* and σ_m^{2*} are set to $\mathcal{N}(0, 10)$, $\mathcal{N}(2, 4)$ and $\Gamma(2, 2)$, respectively. In addition, x_i^* is generated using $x_i^* = F^* z_i^* + \mu^* + \Sigma^*$.

We used the following procedure to conduct the simulation experiment.

- Step 1 The values of z_i , and y_{ij} are generated using $\mathcal{N}(0, \sigma_z^2 I)$ and $\mathcal{N}(z_i z_j^T, \sigma_y^2)$, respectively.
- Step 2 The values of F^* , μ^* and σ_m^{2*} are generated using $\mathcal{N}(0, 10)$, $\mathcal{N}(2, 4)$ and $\Gamma(2, 2)$, respectively.
- Step 3 The value of x_i is generated by $\mathcal{N}(F^* z_i^* + \mu^*, \Sigma^*)$.
- Step 4 The values of z_i , y_{ij} , F , μ and Σ are estimated using the proposed method.
- Step 5 The RMSE between the true value and the estimated parameters is calculated.

In addition to this, note that because of the rotational indeterminacy of F , z_i , we calculated the difference between the true and estimated values of FF^T , $z_i^T z_i$.

Additionally, to confirm the proposed model can capture the true model, we compute the BIC. In this experiment, we set the value of $L = 1$, $d = 5$ and the true dimension of $k = 3$, and calculate the BIC when $k = 1, 2, \dots, 5$.

B. Experimental results

Fig. 2 shows the results of an experiment using the artificial dataset. And Fig. 3 shows the result of the computation of BIC.

It can be seen from Fig. 2 that the proposed method can capture the true value regardless of the value of L . It can also be seen that the proposed method converges after approximately five iterations in all cases. The results for the other parameters, σ_z^2 , σ_y^2 and μ , were similar, although they are not discussed in this paper.

It can be seen from these results that the proposed method can estimate the true value properly, and the parameters can be estimated regardless of the value of L , suggesting that a small value of L is desirable. This suggests that it is desirable to use a relatively low number of computations to estimate the parameters regardless of the value of L . In addition, from Fig. 3 the lowest BIC is when $k = 3$, therefore the true model can extract properly using BIC.

V. EXPERIMENT WITH 20 NEWSGROUPS DATASETS

A. Experimental conditions

The dimensionality of the input data can be reduced by applying the proposed method. Therefore, the proposed method can be used to improve the computational efficiency of high-dimensional data through a dimensionality reduction. To clarify this, we reduced the dimensionality on the 20 Newsgroups dataset and studied the document classification.

To reduce the dimensionality, the inverse map G of the estimated parameters F is calculated using

$$G = (F^T F)^{-1} F. \quad (21)$$

A low-dimensional representation of the input data is derived by multiplying G with the input data.

To confirm the document classification using such low dimensional data, we conducted an experiment on the document classification using the 20 Newsgroups dataset.

We randomly selected 4 out of 20 categories during the experiment as the input data and computed tfidf for those words whose frequency is more than 15. In the proposed method, y_{ij} is computed by taking the inner product of the input data. For these data, the parameters are estimated using the proposed method with the dimensionality k set to $1/2, 1/4, \dots, 1/10$ of the input data, and the dimensionality of the test data is reduced using Eq. (21). Note that, with the proposed method, the category information of the training data is not used to calculate the transformation matrix F or the inverse transformation matrix G .

For classification, we use k -NN as a classifier. The number of neighbors is set to 5, and the cosine measure is used as the distance measure. A principal component analysis, which is a dimensionality reduction method, is used for comparing the classification accuracy.

B. Experimental results

The results of the experiment are shown in Fig. 4.

From Fig. 4, when the reduction of dimensions k is large, the accuracy of the proposed method may be lower than that of the conventional method; however, as the number of reductions in the dimensionality decreases, the accuracy of the proposed method becomes superior. This indicates that the proposed method achieves a better reduction in the dimensionality than the conventional method. In addition, the classification error is almost the same when the dimensionality is reduced by $1/4$ to $1/12$. Thus, it is possible to reduce the classification error while keeping the computational cost low by comparatively reducing the dimensionality, such as $k = d/12$.

VI. CONSIDERATION

This study defined a generative model for estimating the latent structure of the input data under the assumption that the metric between such data is obtained by their inner product, and proposed a method for such an estimation. We also show algorithms to improve the computational efficiency by applying the Woodbury formula. Based on the experimental results derived through artificial data, we demonstrated that the proposed method can estimate the true parameters with a high level of accuracy. In addition, experiments with actual data show that the proposed method can be used as a method for reducing the dimensionality of the high-dimensional datasets. Therefore, the proposed method is effective as a dimensionality reduction method for high-dimensional datasets.

The proposed method can extract the latent structure of the input data and discover latent data characteristics by interpreting the latent variables z_i and obtained the mapping matrix F .

The proposed method assumes that the input data x_i follows $x_i \sim \mathcal{N}(F z_i + \mu, \Sigma)$, i.e., the mean is $F z_i + \mu$ and the variance is σ_m^2 for each element. However, it is straightforward to modify such a model by assuming that all variances are equal, or that $\mu = 0$. Therefore, flexible modeling is possible by changing the definition of the model according to the characteristics of the input data.

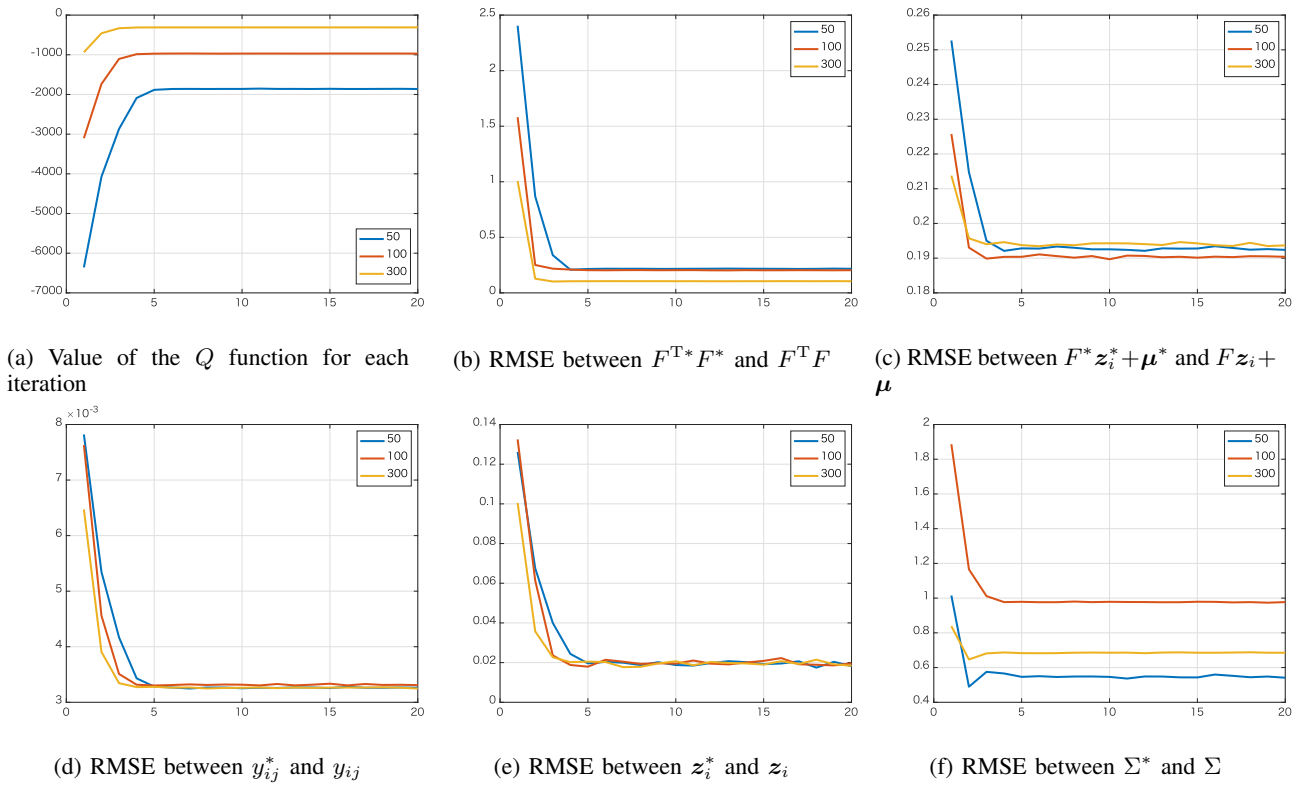


Fig. 2: Experimental results when using artificial data

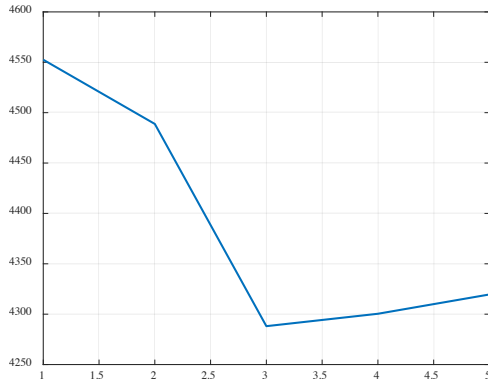


Fig. 3: Experimental results with calculating BIC

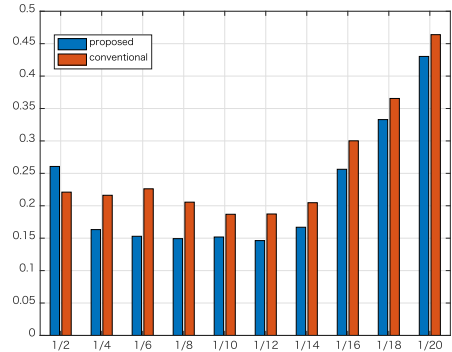


Fig. 4: Experimental results with 20 Newsgroups dataset

VII. CONCLUSION AND FUTURE WORKS

In this study, we proposed a parameter estimation method and an estimation algorithm for efficiently achieving such an estimation under the assumption that the metric between a pair of input data is given based on the latent structure of the given data.

Simulation experiments using artificial data show that the method can appropriately estimate the true value for a given condition. We also showed that our method can achieve a better performance than a conventional dimensionality reduction approach on actual data by leveraging the 20 Newsgroups dataset.

Future studies will include setting an appropriate value of k , assuming a probability distribution appropriate for real data, and deriving an efficient update algorithm under such an assumption.

ACKNOWLEDGEMENT

This work was supported by JSPS KAKENHI Grant Numbers 19K04914 and 21H04600.

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APPENDIX A

DERIVATION OF μ_i AND A_i

To apply the Monte Carlo EM algorithm, we consider the posterior distribution of $\mathbf{Z}$.

$$\begin{aligned} p(\mathbf{Z}|\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}) &= \frac{p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}|\boldsymbol{\theta})}{\int p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}|\boldsymbol{\theta})d\mathbf{Z}} \\ &= \frac{p(\mathbf{X}|\mathbf{Z}, \boldsymbol{\theta})p(\mathbf{Y}|\mathbf{Z}, \boldsymbol{\theta})p(\mathbf{Z}|\boldsymbol{\theta})}{\int p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}|\boldsymbol{\theta})d\mathbf{Z}}. \end{aligned} \quad (22)$$

Here, the numerator of Eq. (22) can express

$$\begin{aligned} &p(\mathbf{X}|\mathbf{Z}, \boldsymbol{\theta})p(\mathbf{Y}|\mathbf{Z}, \boldsymbol{\theta})p(\mathbf{Z}|\boldsymbol{\theta}) \\ &= \prod_{i=1}^n p(\mathbf{z}_i|\sigma_z^2 I)p(\mathbf{x}_i|F, \mathbf{z}_i, \sigma_x^2 I) \left\{ \prod_{j \neq i} p(y_{ij}|\mathbf{z}_i, \mathbf{z}_j, \sigma_y^2) \right\}^{\frac{1}{2}} \\ &= \prod_{i=1}^n \frac{1}{(2\pi)^{\frac{k}{2}}|\sigma_z^2 I|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2\sigma_z^2} \|\mathbf{z}_i\|^2 \right\} \frac{1}{(2\pi)^{\frac{d}{2}}|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \\ &\quad \times \exp \left\{ -\frac{1}{2} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu}) \right\} \\ &\quad \times \left\{ \prod_{j \neq i} \frac{1}{(2\pi\sigma_y^2)^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2\sigma_y^2} (y_{ij} - \mathbf{z}_i^T \mathbf{z}_j)^2 \right\} \right\}^{\frac{1}{2}}. \end{aligned} \quad (23)$$

To focus on the exponential part of Eq. (23), we obtain

$$\begin{aligned} &\exp \left\{ -\frac{1}{2\sigma_z^2} \|\mathbf{z}_i\|^2 \right\} \prod_{j \neq i} \left\{ \exp \left\{ -\frac{1}{2\sigma_y^2} (y_{ij} - \mathbf{z}_i^T \mathbf{z}_j)^2 \right\} \right\}^{1/2} \\ &\quad \times \exp \left\{ -\frac{1}{2} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu}) \right\} \\ &= \exp \left\{ \mathbf{z}_i^T \left(-\frac{1}{2\sigma_z^2} - \frac{1}{2} F^T \boldsymbol{\Sigma}^{-1} F - \frac{1}{4\sigma_y^2} \sum_{j \neq i} \mathbf{z}_j \mathbf{z}_j^T \right) \mathbf{z}_i \right. \\ &\quad \left. - 2 \left(-\frac{1}{2} (\mathbf{x}_i + \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} F + \frac{1}{4\sigma_y^2} \sum_{j \neq i} y_{ij} \mathbf{z}_j^T \right) \mathbf{z}_i + C \right\}, \end{aligned} \quad (24)$$

where C indicates a constant that is independent of the parameters.

From the above discussions, it was shown that the posterior distribution obtained follows a multivariate Gaussian

distribution $\mathcal{N}(\boldsymbol{\eta}_i, A_i)$ with mean $\boldsymbol{\eta}_i$ and variance-covariance matrix A_i . Here, A_i and $\boldsymbol{\eta}_i$ are given by

$$\begin{aligned} A_i &= \left\{ \frac{1}{\sigma_z^2} I + \frac{1}{2\sigma_y^2} \sum_{j \neq i} \mathbf{z}_j \mathbf{z}_j^T + F^T \boldsymbol{\Sigma}^{-1} F \right\}, \\ \boldsymbol{\eta}_i &= A_i^{-1} \left(F^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i + \boldsymbol{\mu}) + \frac{1}{\sigma_y^2} \sum_{j \neq i} y_{ij} \mathbf{z}_j \right). \end{aligned}$$

APPENDIX B

DERIVATION OF INVERSE MAP G OF F

Here, from the assumption that $\mathbf{x}_i = F\mathbf{z}_i + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_i$, let the estimated $F, \mathbf{z}_i, \boldsymbol{\mu}$ and $\mathbf{x}_i$ be $\hat{F}, \hat{\mathbf{z}}_i$ and $\hat{\boldsymbol{\mu}}, \hat{\mathbf{x}}_i$, and thus $\hat{\mathbf{x}}_i = \hat{F}\hat{\mathbf{z}}_i + \hat{\boldsymbol{\mu}}$. Solving this for $\hat{\mathbf{z}}_i$, we obtain the following:

$$\hat{\mathbf{z}}_i = \left(\hat{F}^T \hat{F} \right)^{-1} \hat{F}^T (\hat{\mathbf{x}}_i - \hat{\boldsymbol{\mu}}). \quad (25)$$

Consequently, letting the inverse map of F be $\left(\hat{F}^T \hat{F} \right)^{-1} \hat{F}^T$, the reduction in the dimensionality can be achieved by subtracting the estimated $\boldsymbol{\mu}$ from the input data and multiplying the result by G .